SMART AGRI-DOCTOR

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Summary

This report presents a detailed software metrics analysis of **Smart Agri Doctor**, a mobile-based intelligent agriculture support system developed to assist farmers in identifying crop diseases and receiving treatment recommendations. The application leverages artificial intelligence and image processing techniques to analyze crop images and provide accurate disease detection results. The system is designed to be user-friendly, scalable, and maintainable, ensuring practical usability in real agricultural environments.

Project Type: Final Year Project (FYP)
Domain: Artificial Intelligence & Smart Agriculture
Technology Stack: Flutter (Dart), TensorFlow Lite, Image Processing Libraries, Weather API

1 Introduction

Smart Agri Doctor addresses the growing need for intelligent agricultural solutions by providing farmers with a digital platform for crop health analysis. The system enables users to upload crop images, detect diseases using AI models, and receive recommendations for preventive and corrective measures. This reduces dependency on manual inspection and improves crop productivity.

1.1 Project Objective

- Detect crop diseases using image-based analysis
- Provide AI-driven recommendations for disease treatment
- Offer a simple and accessible mobile interface for farmers
- Ensure system scalability and maintainability

1.2 Scope of Project

- Crop image upload and processing
- Disease detection using trained AI models
- Display of disease details and recommendations
- User interaction through a mobile application

2 Product Metrics

2.1 Lines of Code (LOC)

Static analysis of the complete project source code resulted in the following measurements:

Metric	Count
Total Lines of Code (LOC)	1,148
Source Lines of Code (SLOC)	824
Logical Lines of Code (LLOC)	612
Comment Lines	104
Blank Lines	220

Comment Percentage:

$$C\%LOC = \frac{104}{1128} \times 100 = 9.2\%$$

2.2 Function Point Analysis

Components	Count
External Inputs (EI)	6
External Outputs (EO)	7
External Inquiries (EQ)	6
Internal Logical Files (ILF)	2
External Logical Files (EIF)	2

Detailed Explanation:

External Inputs (EI = 6):
1. Crop image upload
2. User login input
3. Crop selection input
4. Disease analysis request
5. Feedback submission
6. Model initialization

External Outputs (EO = 7):
1. Disease prediction result
2. Confidence score
3. Treatment recommendation
4. Error messages
5. Analysis status output
6. Crop health summary
7. Notification output

External Inquiries (EQ = 6):
1. Disease information query
2. Crop details query
3. Recommendation query
4. Analysis history query
5. Model status query
6. App status inquiry

Internal Logical Files (ILF = 2):
1. Trained disease detection model
2. User session data

External Logical Files (EIF = 2):
1. Image dataset files
2. External configuration files

2.3 Cyclomatic Complexity Metrics

Module	Complexity
Image analysis module	10
Disease classification	8

Recommendation engine	5
UI navigation logic	4
Data handling	3

Total Complexity: 30

Average Cyclomatic Complexity:

$$\frac{30}{5} = 6$$

Risk Level: Moderate

2.4 Halstead Complexity Metrics

Observed Data:

Metric	Value
Unique Operators (h1)	14
Unique Operands (h2)	132
Total Operators (N1)	136
Total Operands (N2)	248

Halstead Calculations:

- Vocabulary (h) = h1 + h2 = 146
- Length (N) = N1 + N2 = 384
- Volume (V) = N × $\log_2$(h) = 2984.12
- Difficulty (D) = (h1 / 2) × (N1 / h2) = 12.32
- Effort (E) = V × D = 36760.6
- Time to Implement = E / 18 = 34.04 minutes
- Estimated Bugs = V / 3000 = 0.99

2.5 Maintainability Index

$$MI = 171 - 5.2\ln(2984.12) - 0.23(6) - 16.2\ln(824)$$

Maintainability Index = 18.6

Maintainability Level: Moderately Maintainable

2.6 Quality Metrics

2.6.1 Defect Density

- **Total Defects Identified**: 290
- **KLOC** = 824 / 1000 = 0.824

$$DefectDensity = \frac{290}{0.824} = 351.9 \text{ defects/KLOC}$$

2.6.2 Test Coverage

Test No	Module
1	Image upload
2	Disease detection
3	Recommendation system
4	User authentication

5	Data storage
6	Error handling

3 Process Metrics

3.1 Development Time (COCOMO Model)

Project Type: Semi-detached

SLOC = 824

KLOC = 0.824

$$Effort = 3.0 \times (0.824)^{1.12} = 2.45 \text{ Person-Months}$$

3.2 Development Duration

$$Time = 2.5 \times (2.45)^{0.35}$$

Estimated Development Time: 3.2 Months

3.3 Cost Estimation

Assuming **PKR 80,000 per person-month**:

$$Cost = 2.45 \times 80{,}000 = PKR196{,}000$$

3.4 Feature Point Analysis

Component	Count	Weight	Result
EI	6	4	24
EO	7	5	35
EQ	6	4	24
ILF	2	10	20
EIF	2	7	14

Unadjusted FP (UFP): 117
Complexity Factor ∑fi = 32

$$FP = 117(0.65 + 0.01 \times 32) = 132FP$$

Productivity:

$$132/2.45 = 53.8FP/PM$$

Cost per Function:

$$196{,}000/53.8 = PKR3645 \text{ per function}$$

3.5 Defect Removal Efficiency

- Total Defects = 290
- Total Feature Points = 117

$$DefectDensity = 290/117 = 2.47$$

$$DRE = \frac{120}{290 + 120} \times 100 = 29.2\%$$

3.6 Cycle Time & Throughput

Module	Cycle Time (Days)
Dataset preparation	12
Model training	16
App development	14
Integration	10
Testing	14

Total Cycle Time: 66 days
Average Cycle Time:

$$66/5 = 13.2 \text{ days}$$

Throughput:

$$5/2.2 = 2.27 \text{ modules/month}$$

4 Tools & Technology

Tool	Purpose
Flutter (Dart)	Mobile UI development
Firebase	Database & authentication
Python	Backend processing
Machine Learning	Disease detection
Image Processing	Crop image analysis

5 Problems & Challenges

- Limited quality of crop images
- Variations in lighting conditions
- Dataset imbalance
- Model accuracy for rare diseases
- Internet dependency in rural areas

6 Conclusion

Smart Agri Doctor provides an effective AI-based solution for disease detection and crop health analysis. The software metrics evaluation indicates that the system is moderately maintainable, scalable, and suitable for real-world agricultural applications. The project successfully achieves its objectives and offers a strong foundation for future improvements, such as multilingual support and offline analysis.